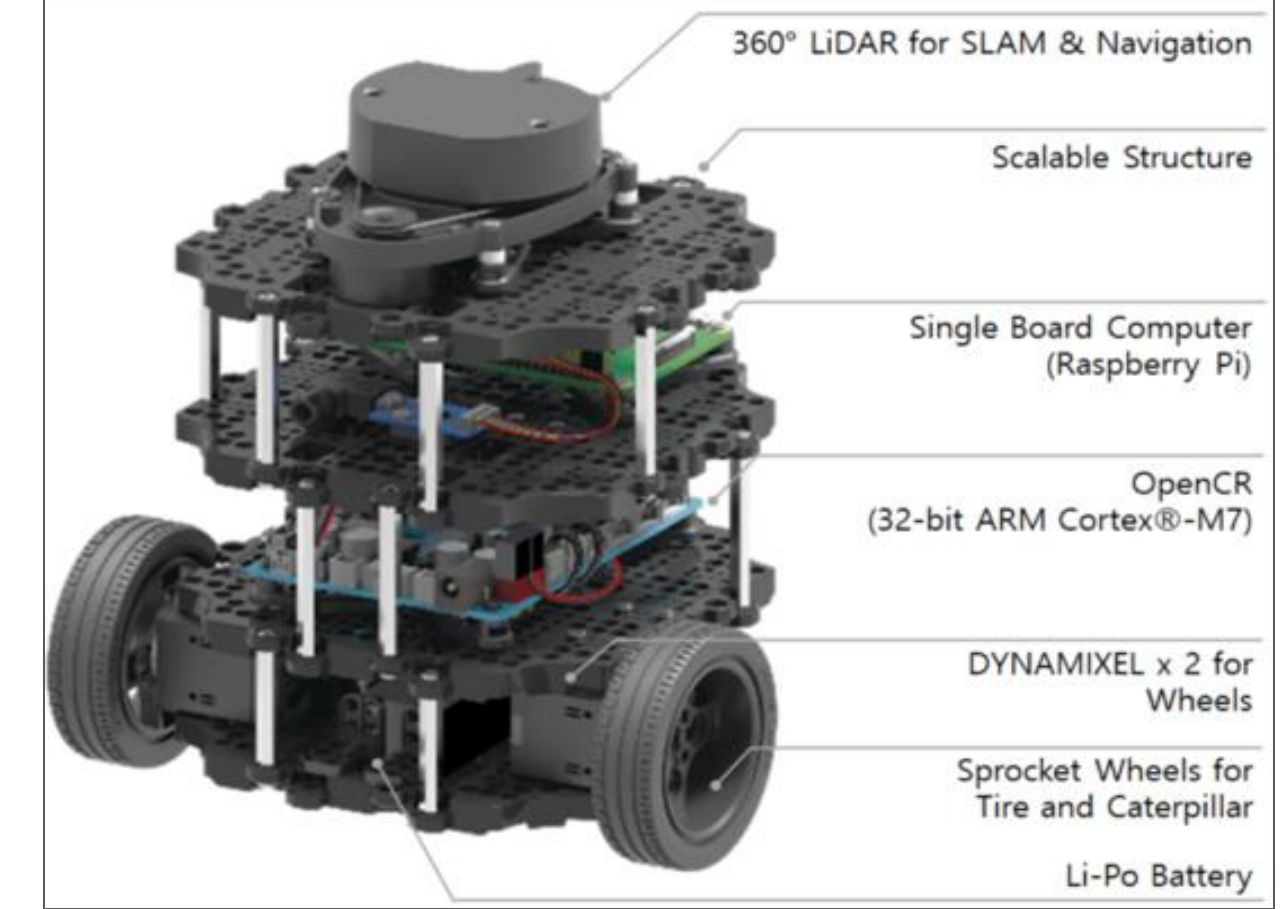


# A Comparative Study of SLAM Algorithms for Robot Operating System

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## MOTIVATION

- Simultaneous Localization and Mapping (SLAM) is a useful algorithm for robot navigation and mapping of the environment at the same time
- Improving SLAM precision supports advanced applications such as autonomous exploration and loop-closure correction in dynamic environments
- Two popular SLAM algorithms, SLAM Toolbox and Cartographer produce different quality map under identical conditions, affecting localization and navigation

## BACKGROUND/PRIOR WORKS

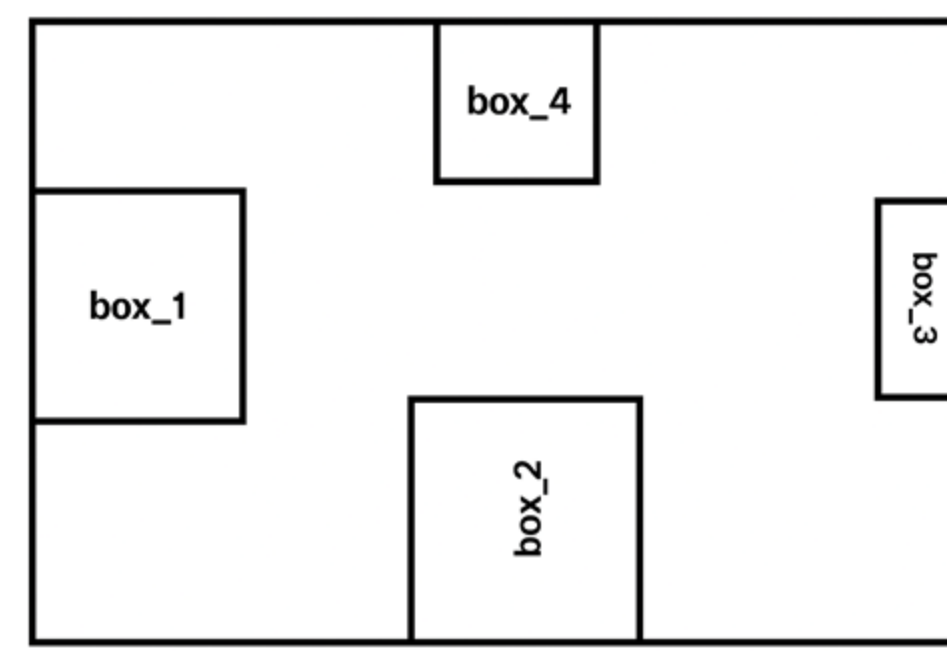
- Li & Zhu (2024)** evaluated three 2D LiDAR SLAM algorithms Hector, GMapping, and Cartographer in simulated orchard environments using Gazebo
  - Cartographer achieved the best localization and mapping accuracy ( $\approx 8$  cm error at 4 cm resolution) but required more CPU and memory
  - Performance metrics: Localization error RMSE, Mapping error MAE, CPU and memory usage

## PROBLEM FORMULATION

- This study compares Cartographer and SLAM Toolbox to determine which SLAM algorithm generates more accurate and consistent maps under identical conditions
- Although both are widely used for autonomous navigation and mapping, performance differences arise from sensor noise, environment geometry, and algorithm design
- A controlled maze environment with fixed ground-truth dimensions was used to ensure fair testing and repeatable conditions



Test Environment

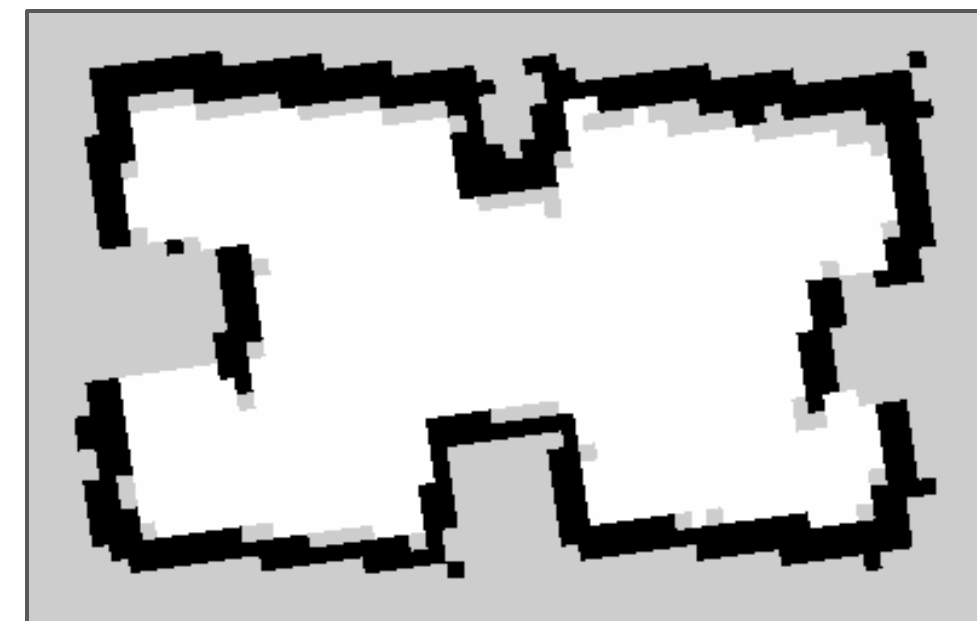


Model Environment

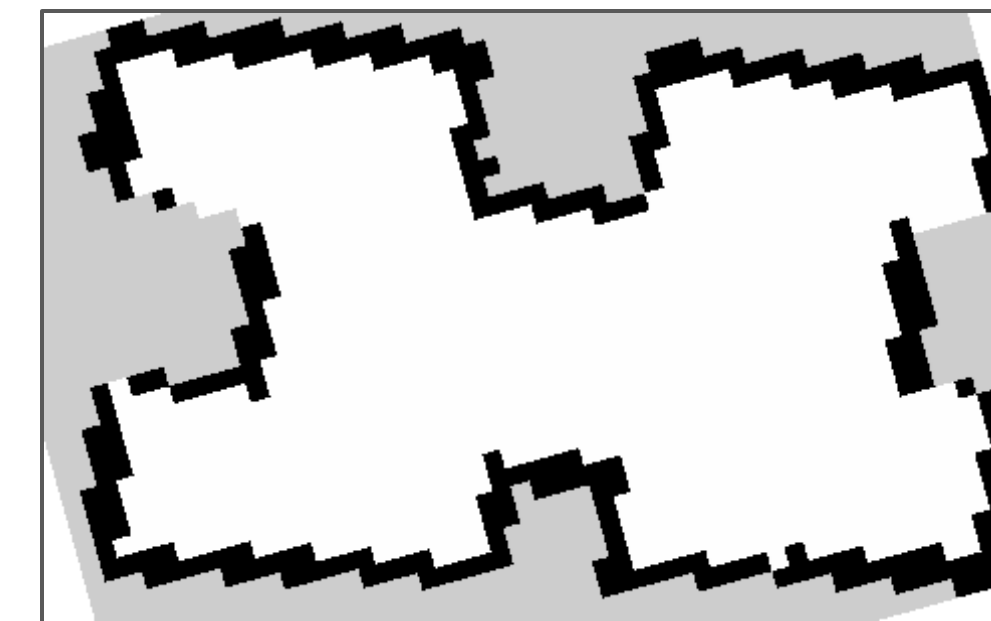
- The experiment quantifies mapping accuracy, reliability, and structural fidelity, identifying which algorithm provides superior geometric precision for the TurtleBot 3 platform

## METHODOLOGY

- Both algorithms were configured using the default software setup from the original authors [1, 2]
- Using each algorithm, the real TurtleBot was driven over predefined path at the speed of 0.01 m/s
- Tests were performed 10 times in the same maze environment to ensure repeatability and gauge variance for each algorithm

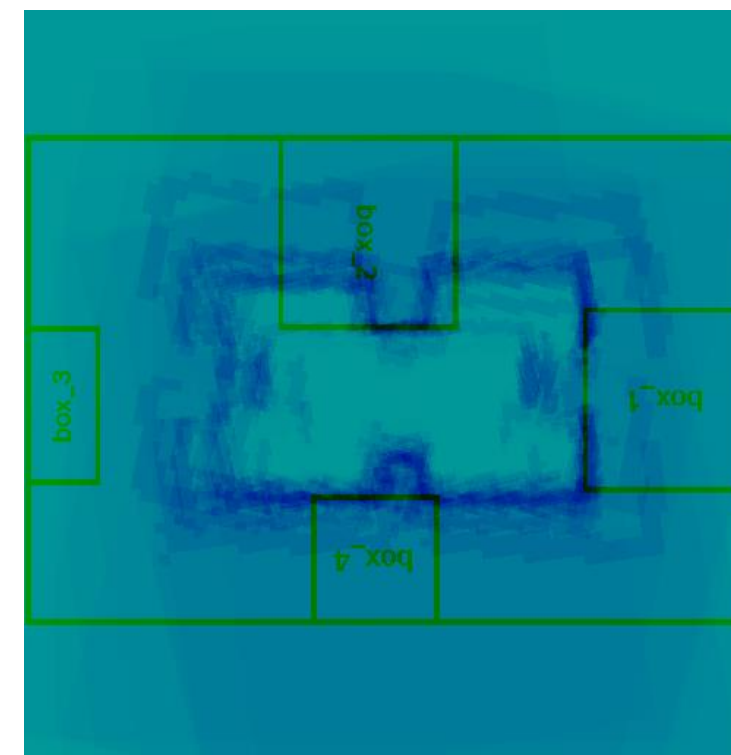


Cartographer

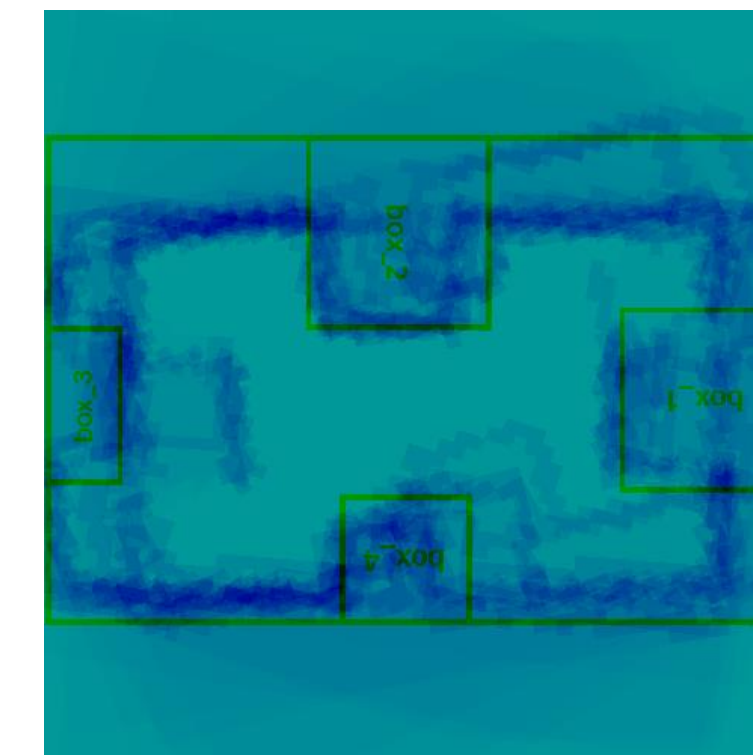


SLAM Toolbox

- All saved maps were resized to a fixed pixel dimension (500 by 500 pixels) to maintain consistent scaling
- A Python script was developed to overlay maps and visually align them for direct qualitative comparison



Cartographer



SLAM Toolbox

- The program also computed the map error by quantifying pixel-level deviations between the generated SLAM maps
- Measures the average Euclidean distance between each map point and its corresponding ground-truth point
- The mean squared error and variance were calculated and map-error values obtained across all runs, providing a measure of overall consistency and spread

$$e_M = \frac{1}{N} \sum_{i=1}^N \sqrt{(x_i - x_i^{GT})^2 + (y_i - y_i^{GT})^2}$$

## RESULTS

| Metric                      | SLAM Toolbox | Cartographer |
|-----------------------------|--------------|--------------|
| Mean Error (px)             | 19.294       | 25.227       |
| Mean Error (cm)             | 96.468       | 126.134      |
| MSE (px <sup>2</sup> )      | 379.448      | 641.909      |
| Variance (px <sup>2</sup> ) | 7.207        | 5.517        |
| Correlation                 | 0.169        | 0.246        |

- SLAM Toolbox has a lower error and a lower MSE proving more accuracy
- Cartographer has a lower variance and higher correlation proving it is slightly more consistent
- The combination of metrics indicates that SLAM Toolbox maps were closer to the real environment, Cartographer maps were marginally more uniform in shape

## DISCUSSION

- TurtleBot3 with ROS2 provide an ideal testbed for evaluating SLAM algorithm accuracy and reliability in real-world robotics
- The lower average map error produced by SLAM Toolbox means the robot's perceived environment was physically closer to the true maze geometry
- SLAM Toolbox's performance suggests stronger handling of local sensor noise and small-scale geometric deviations
- Cartographer may smooth map features more aggressively, reducing noise but also increasing systematic drift in constrained environments like the maze

## REFERENCES

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